

Saahil Mohal

Atlanta | D.C. | (703) 996-6908 | saahil.mohal@gatech.edu | saahilmohal.com | US Citizen



EDUCATION

Georgia Institute of Technology

Fall 2023 - Summer 2027

M.S. in Mechanical Engineering | Expected Summer 2027

B.S. in Mechanical Engineering | Expected Fall 2026

GPA: 3.90 - Faculty Honors (3x), Dean's List (2x)

Relevant Coursework: Robotics, Computer Vision, ML, Modeling & Sim, Machine Design, Heat Transfer, System Dynamics

Organizations & Activities: Woodruff School Student Advisory Council, ElectrifyGT (Project Lead), Interfraternity Council (VP of Communications, Technology Director), Delta Chi Fraternity (Judicial Board, Scholarship Chair)

WORK EXPERIENCE

Odin Dynamics

Los Angeles, CA

June 2026 - August 2026

Electro-Mechanical Integration Intern

- Led end-to-end design of a deployable Ku-band satellite-communications subsystem for an unmanned underwater vehicle (UUV), owning requirements, and MIL-STD selection: authored Python RF models optimizing a fused-quartz radome composite layout to <0.3 dB insertion loss across $\pm 55^\circ$ scan; re-architected a flat window failing at 300 m depth ($10\text{--}30\times$ allowable σ) into a streamlined curved membrane-compression geometry (>8 FoS).
- Designed, procured, and assembled an underwater thruster dynamometer rated to 2,500 N / 160 N·m through a passed Critical Design Review: authored a 10-failure-mode structural package (FoS ≥ 6), built a Python beam-element model from CAD for unmeshable components, cut deflection 85% with diagonal bracing, and mitigated galvanic corrosion at aluminum-stainless interfaces.
- Led the field campaign qualifying Starlink for submerged maritime use: built a Python gRPC harness logging 44 telemetry channels at 1 Hz and devised mass-based water-depth metrology (0.008 mm) — proving sub-millimeter water films cause 82% packet loss, pivoting to recovery-time testing across hydrophobic coatings and mount angles.
- Architected a kW-scale floating-HV power distribution board converting a 108–151 V pack to an isolated 48 V bus with point-of-load rails ($\sim 16\times$ lower distribution loss), EV-grade pre-charge/contactor protection with BMS-gated fail-safe sequencing, and per-rail eFuse telemetry to an NVIDIA Jetson over isolated CAN.
- Designed and procured a four-link RF front end (5G 4x4 MIMO, precision GNSS, Iridium satcom, Wi-Fi) for the UUV's surface antenna, resolving a ~ 150 dB co-site dynamic-range conflict between GNSS receive and cellular transmit with a specified Iridium-reject cavity notch filter.

Lockheed Martin Co.

Orlando, FL

June 2025 - August 2025

Mechanical Engineering Intern | Rotary & Mission Systems (RMS) + Missiles & Fire Control (MFC)

- Engineered a Python computer vision pipeline using color segmentation to quantify coating delamination during high-velocity rain erosion testing. Leveraged this data to qualify a new F-35 EOTS frame material, reducing unit cost by 78% ($>\$70k/yr$ savings) while maintaining RCS compliance.

CACI International Inc.

Sterling, VA

May 2024 - August 2024

Mechanical Engineering Intern | National Security & Innovative Solutions (NSIS)

- Designed and prototyped a field-ready GPS antenna characterization device, integrating a precision-machined PCB substrate, conducting conjugate heat transfer analyses, and managing BOM procurement.

PROJECTS

Automated Pet Feeding System | Personal Project

August 2025 - January 2026

- Designed and fabricated a custom 2-layer PCB (KiCad) featuring dual buck converters to regulate 20V input; integrated logic-level N-Channel MOSFETs for high-current switching with flyback diodes and decoupling capacitors to isolate sensitive logic from inductive motor noise.

Eagle Scout | Former Senior Patrol Leader of BSA Troop 1159

August 2016 - December 2022

SKILLS

- CAD/Simulation:** SolidWorks, Siemens NX, CATIA, Ansys (FEA/CFD), Additive/Traditional Mfg Methods
- Programming & Data:** Python (NumPy, Pandas, OpenCV, gRPC), MATLAB, Simulink, Git, Linux
- Hardware & Controls:** PCB Design (KiCad), Signal Conditioning, Op Amps, PWM, DAQ & Load Cell Integration, Oscilloscopes, S-Parameter/RF Analysis, Arduino, Raspberry Pi